

Chapter 5 In-Class Exercise: Searching a Matrix

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1 Matrices Using Lists

1. A two-dimensional array of values can be represented in Python using a **list of lists**. Each inner list represents one row. Begin with the matrix

$$A = \begin{bmatrix} 4 & 7 & 2 & 9 \\ 1 & 5 & 8 & 3 \\ 6 & 0 & 4 & 7 \end{bmatrix}.$$

Represent this matrix in Python using

```
A = [
    [4, 7, 2, 9],
    [1, 5, 8, 3],
    [6, 0, 4, 7]
]
```

- The first index selects a row and the second index selects a column. Try

```
print(A[0][0])
print(A[1][2])
print(A[2][3])
```

- Remember that Python begins counting at zero. Therefore, `A[1][2]` is the element in the second row and third column.

2 Nested Loops

1. A loop inside another loop is called a **nested loop**. The outer loop can move through the rows, while the inner loop moves through the columns. Use

```
for row in range(3):
    for column in range(4):
        print(A[row][column])
```

- How many times does the outer loop execute?
- How many times does the inner loop execute?
- How many total matrix elements are printed?
- Modify the print statement to display the row, column, and value:

```
print(row, column,
      A[row][column])
```

3 Searching for a Value

1. Now search the matrix for a particular value. Begin with

```
target = 7
```

Use nested loops and an `if` statement:

```
for row in range(3):
    for column in range(4):

        if A[row][column] == target:
            print("Found at",
                  row, column)
```

- Run the program. How many times does the value 7 occur?
- At which row and column indices does it occur?
- Change

```
target = 7
```

to several other values and repeat the search.

4 In-Class Exercise

1. Create the matrix

$$B = \begin{bmatrix} 12 & 5 & 8 & 3 & 11 \\ 7 & 14 & 2 & 9 & 6 \\ 4 & 10 & 13 & 5 & 1 \\ 8 & 3 & 6 & 12 & 7 \end{bmatrix}.$$

Represent it as a list of four lists.

- Ask the user for an integer to search for:

```
target = int(input(
    "Value to search for: "))
```

- Use nested `for` loops to search every row and column.
- Whenever the value is found, print

```
Value found at row __ column __
```

using the actual row and column indices.

- Create the Boolean variable

```
found = False
```

before beginning the search. Whenever a match occurs, change it to

```
found = True
```

- After both loops have finished, use

```
if not found:
    print("Value not found")
```

so that the program also correctly handles values that are not inside the matrix.

- Test your program using one value that occurs once, one value that occurs more than once, and one value that does not occur.